



SANA MOIN

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English (C1), German (B1)

STRENGTHS

- Experienced with agile development and best practices, with a focus on problem-solving and innovation to find actionable insights from data.
- Proficient in data science and machine learning concepts, algorithms, and statistical analysis techniques. Experience in developing and deploying machine learning models and working with large datasets using Python and SQL.
- Effective communication and collaboration skills to convey complex technical information to non-technical stakeholders.
- Skilled in multiple programming languages and working across different technology stacks, with attention to deliver high-quality code that meets user needs.

SKILLS

- Machine Learning	- Deep Learning	- Speech Processing	- Computer Vision	- Natural Language Processing	- Linux
- PyTorch	- Python	- Robotics Basics	- Docker	- Cryptography basics	- SQL

EXPERIENCE

→ Research Assistant

Fraunhofer CML, Hamburg

Jul 2021 - Present

- Developed and deployed an **automatic speech recognition** system using transformer models and applied audio preprocessing techniques to noisy maritime data, achieving a 20% reduction in word error rate.
- Implemented **PyTorch Lightning** and **DDP** approaches to train a large-scale ASR model on extensive speech datasets, resulting in improved accuracy and faster training times.
- Doubled the performance of the ASR model by **generating synthetic maritime data** to develop a language model.
- Led the **design of the ASR software architecture** and conducted research in a cross-functional team to deploy it to the customer, enabling near-real-time automatic transcription.

→ Software Engineer

Philips, Bengaluru, India

Jul 2018 - Oct 2020

- Collaborated with cross-functional teams, including databases, DevOps, deployment, and customer-facing teams to develop a **software system for performing ETLs** of hospital medical data using JavaScript, Java, and SQL in a scrum team.
- Provided production-level upgrades, enhancements, and urgent bug resolutions with a proven track record of independently **resolving 60% of the team's bugs**, also getting recognized for ownership and fast delivery.
- Led the **code refactoring** effort, eliminating over 600 code smells and security vulnerabilities, achieving a **30% improvement** in code performance and reducing over 400 lines of code.
- Developed an innovative **functional testing tool** that earned recognition and resulted in 80% reduction in testing efforts.
- Conducted **knowledge-sharing sessions** and was recognized for eagerness to inspire and improve the team's skills.

→ Research intern

Philips Research, Bengaluru, India

Jan 2018 - Jun 2018

- Performed ETL of medical text data and implemented a NER system using TensorFlow to automatically **de-identify Protected Health Information** in free-text medical records

EDUCATION

→ MSc. Intelligent Adaptive Systems

Universität Hamburg

Nov 2020 - Present

Thesis: Federated learning with homomorphic encryption for medical data on neural networks

Key Projects:

- Collaborative indoor Multi-Agent Navigation with AI2-THOR simulator and CLIP for visual+textual embedding with A3C reinforcement learning.
- Meme offensive content detection from images with ensemble of computer vision, BERT, ML and bi-LSTM with contextual information
- NLP Relation Extraction for three Low Resource Indian Languages using mBERT and multilingual RoBERTa model
- Predict hate/offensive speech in movies using BERT/Bi-LSTM and introduced novel dataset of 11k movie subtitles

→ B.Tech. Computer Science and Engineering,

MIT, Manipal, India

Jul 2014 - May 2018

PUBLICATIONS

- "How Hateful are Movies? A Study and Prediction on Movie Subtitles"-Proceedings of the 17th Conference on NLP, KONVENS 2021
- "De-Identification of Protected Health Information PHI from Free Text in Medical Records"-IJSPTM Vol 8, No 1/2, May 2019
- "The Performance Enhancement of Statistically Significant Bicluster Using Analysis of Variance"-Konkani A., Bera R., Paul S. (eds) Advances in Systems, Control and Automation. Lecture Notes in Electrical Engineering, vol 442. Springer, Singapore, 2017

EXTRA CURRICULAR

- Received four recognitions at Philips
- Philips hackathon project finalist.
- Elected as Class representative during bachelor for 2016-18